Model

A Algorithm

$$G_t = G(x_t, \xi_t),$$
$$x_{t+1} = \arg\min_{x \in X} \{\gamma_t \langle G_t, x \rangle + V(x_t, x)\}.$$

T Main theorem

$$\mathbb{E}[f(\bar{x}_s^k)] - f^* \le \left(\sum_{t=s}^k \gamma_t\right)^{-1} \left[\mathbb{E}[V(x_s, x^*)] + (M^2 + \sigma^2) \sum_{t=s}^k \gamma_t^2\right].$$

P Source proof

$$\delta_t := G(x_t, \xi_t) - g(x_t),$$
$$\mathbb{E}[\delta_t \mid \mathcal{F}_{t-1}] = 0.$$

Lean model

```
structure
  MirrorDescentProcess
  ... where
  ξ : Ω →
    {ξ : SampleSpace d //
      ξ ∈ setup.Ξ}
  ξt :
    {t : ℕ // 1 ≤ t} →
    Ω →
    {ξ : SampleSpace d //
      ξ ∈ setup.Ξ}
  hξt_measurable :
    ∀ t, Measurable (ξt t)
  ...
```

Reuse

Prove

Target theorem $T^\star$

Lean goal

```
⊢ volume[δ | F]
    =ᵐ[volume] 0
```

Blueprint

Proof obligations

- Descent step ✓
- **Conditional expectation**
- Measurability
- Output bound

Planner

Prover

Proof trajectory

➤ Current Lean goal

```
let F := proc.filtration.seq (t.1 - 1)
let δ := fun ω =>
  proc.G t ω - proc.g (proc.x t ω)
⊢ volume[δ | F] =ᵐ[volume] 0
```

➤ Retrieve a mean-zero lemma

$\mathbb{E}[\varepsilon(x)] = 0$ for each **fixed x**.

➤ Try it at the current iterate x

x depends on F. The conditional goal is **still open**.

➤ Inspect how x and ξ depend on history F

- Show x is **F-measurable**: it uses only the past.

```
theorem x_past_measurable
    (t : MirrorDescentSetup.Time) :
    Measurable[proc.filtration.seq (t.1 - 1)]
      (proc.x t) := by ...
```

- Show the current sample ξ is **independent of F**.

Audit

Completed proof

Reconstruct

Expose sampling history

revise

Signature contract

Source: i.i.d. sampling
Scope: stochastic process

```
hξt_iIndep : iIndepFun ξt volume
hξt_identDistrib :
  ∀ t, IdentDistrib (ξt t) ξ volume volume
```

Role: derived lemma

```
theorem
  oracle_noise_condExp_zero_of_iid_adapted
    (t : MirrorDescentSetup.Time) :
    volume[
      (fun ω => proc.G t ω
        - proc.g (proc.x t ω)) |
      proc.filtration.seq (t.1 - 1)
    ] =ᵐ[volume] 0 := by ...
```

Judge

① Scope check: Source-backed process change?

② Claim check: Same algorithm and theorem?

③ Second Judge: Agrees with first Judge?

approve → `MirrorDescentProcess`

Resume proving

Certify

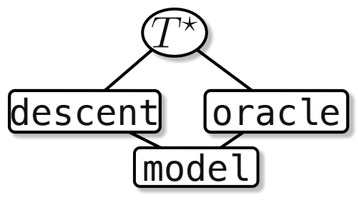
Lean

$\vdash T^\star$

✓ No goals

source audited

Extract and Merge



Search & inspect

SOptLib

- Glue/
  - Probability.lean
- Model/
  - Filtration.lean
- Layer0/
  - **Oracle.lean**
- Layer1/
  - Descent.lean

Layer0/Oracle.lean

```
theorem
  condExp_oracle_noise_eq_zero_of_iid_adapted
    ... :
    μ[(fun ω => G (x ω) (Y ω) - g (x ω)) | m]
      =ᵐ[μ] 0 := by ...
```

Construction records

knowledge/

- **Modeling**
  - Sampling & freshness
- Proof patterns
- Pitfalls
- Conventions